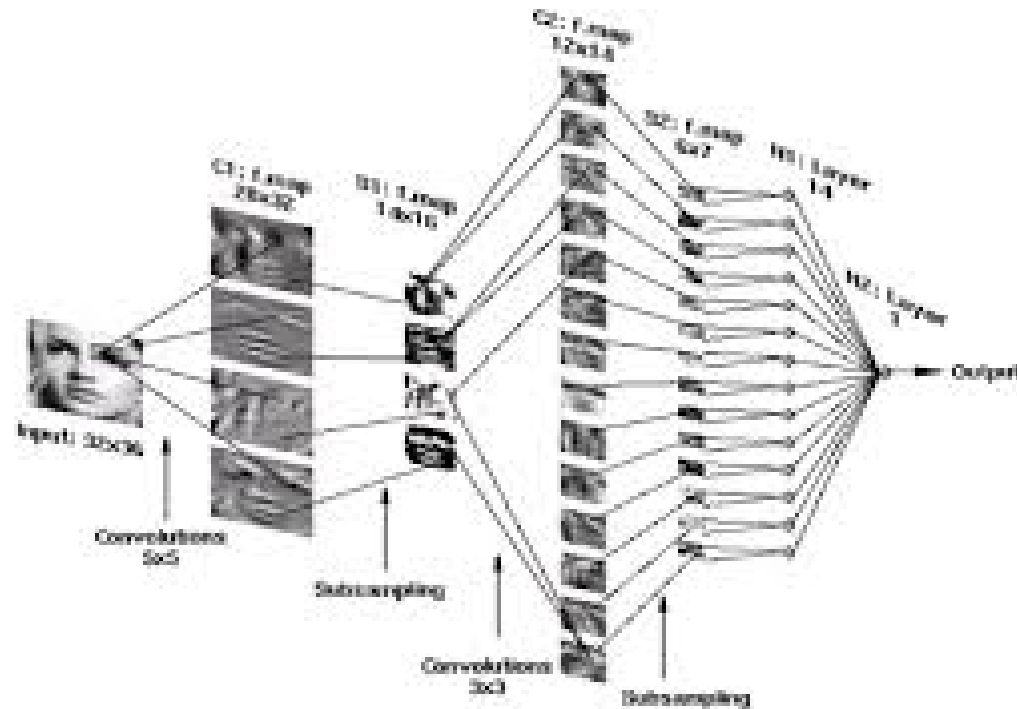


Mathematics of Machine Learning

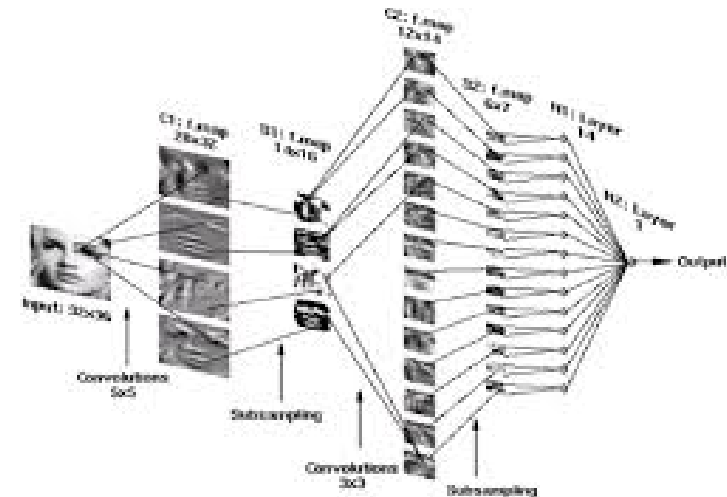
Introduction



Course Details and Topics

Outline

1. **Why are you doing this course?**
2. What we assume that you know



Orientation

- This course addresses three cohorts of students:
 - ★ Third year Artificial Intelligence Students
 - ★ MSc in Machine Learning Students
 - ★ Volunteers
- It is dual coded (AICE3001/AICE6002), with subtle difference in expectations reflected in slightly different evaluation
- Notes are on Moodle and

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- LLMs can help us a lot, but if you don't trust yourself you are disempowered
- This course is aimed at building up trust in your understanding of how machine learning works

Why Mathematics?

- Learning machines are notoriously difficult to understand
- Mathematics allows us to reason rigorously
- Posing question mathematically and seeking mathematical solutions provide allow us to build a strong intuition
- Mathematics also allow us to describe the world precisely
- It also provides an arsenal of tools that turn out to be hugely useful

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Cracking the Code

- Mathematics is a language that takes time to master
- For some it is easy, for others it will be a struggle
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Starting from Different Places

- You all start from different places
- Some of you may have a degree in mathematics
- For others it may have been your least favourite subject
- I invite strong mathematicians to become even stronger
- For students starting from a weak base, judge yourselves by your progress, not by comparing yourself to others

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- Understand every step
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The Journey

- We are going to visit lots of aspects of mathematics that have proved fruitful in machine learning
- Some of this will be very concrete, while in other cases by look at a higher level of abstraction we can generalise methods across different application areas
- We will cover, calculus, vector spaces, convexity, optimisation, entropy, probability, statistics and much else besides
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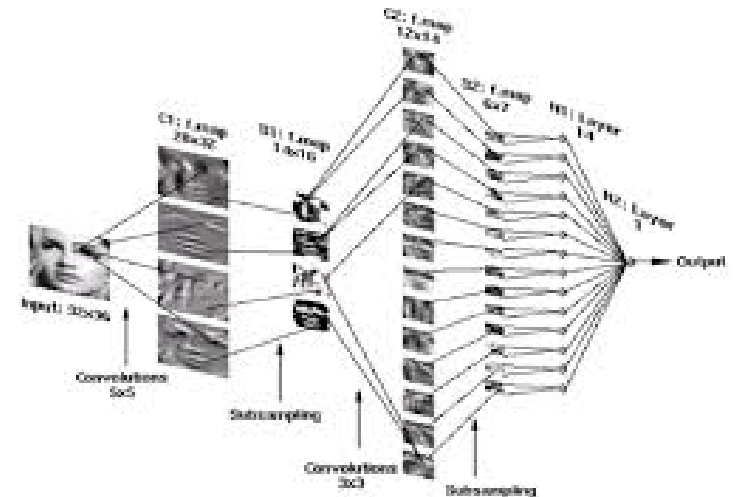
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What we Assume you Know

- You should all have A-level mathematics and have done some mathematics at Uni
- You should know what a function is, e.g. $f(x) = x^2 + 3$
- You should be able to do simple algebraic manipulations
- And some elementary calculus

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Vectors

- Ordinary vectors (what mathematicians call elements of $\mathbb{R}^n$) are much used in machine learning and should be familiar
- We will use bold symbols to represent **column vectors**, e.g.

$$\mathbf{x} = \begin{pmatrix} 1.2 \\ 2.3 \\ 4.7 \end{pmatrix}$$

- We denote **row vectors** as the **transpose** of column vectors
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- We care whether vectors are row or column vectors as it allows us to write equations involving vectors succinctly and unambiguously

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Matrices

- Matrices are blocks of number that define linear mappings between vectors, e.g. $\mathbf{M} \in \mathbb{R}^{2 \times 3}$

$$\mathbf{M} = \begin{pmatrix} 1.1 & -2.1 & 6.2 \\ 9.3 & 3.1 & -4.4 \end{pmatrix}$$

- An $m \times n$ matrix (i.e. $\mathbf{M} \in \mathbb{R}^{m \times n}$) has m rows and n columns
- An $m \times n$ matrix acting on a vector in $\mathbb{R}^n$ transforms it to $\mathbb{R}^m$ vector, e.g.

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Composition of Mappings

- Matrices, $\mathbf{A}$, $\mathbf{B}$, etc. represent mappings between spaces
- Provided the matrices are the correct size we can compose these mappings, e.g. if $\mathbf{A} \in \mathbb{R}^{\ell \times m}$ and $\mathbf{B} \in \mathbb{R}^{m \times n}$ then $\mathbf{AB}$ represents the effect of
 - ★ first applying $\mathbf{B}$ to vectors in $\mathbb{R}^n$ to obtain vectors in $\mathbb{R}^m$
 - ★ and then applying $\mathbf{A}$ to get vectors in $\mathbb{R}^\ell$

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Matrix Multiplication

- The combined mapping $\mathbf{AB}$ can be represented by an $\ell \times n$ matrix $\mathbf{C}$ which we can compute using matrix multiplication (rows times columns)

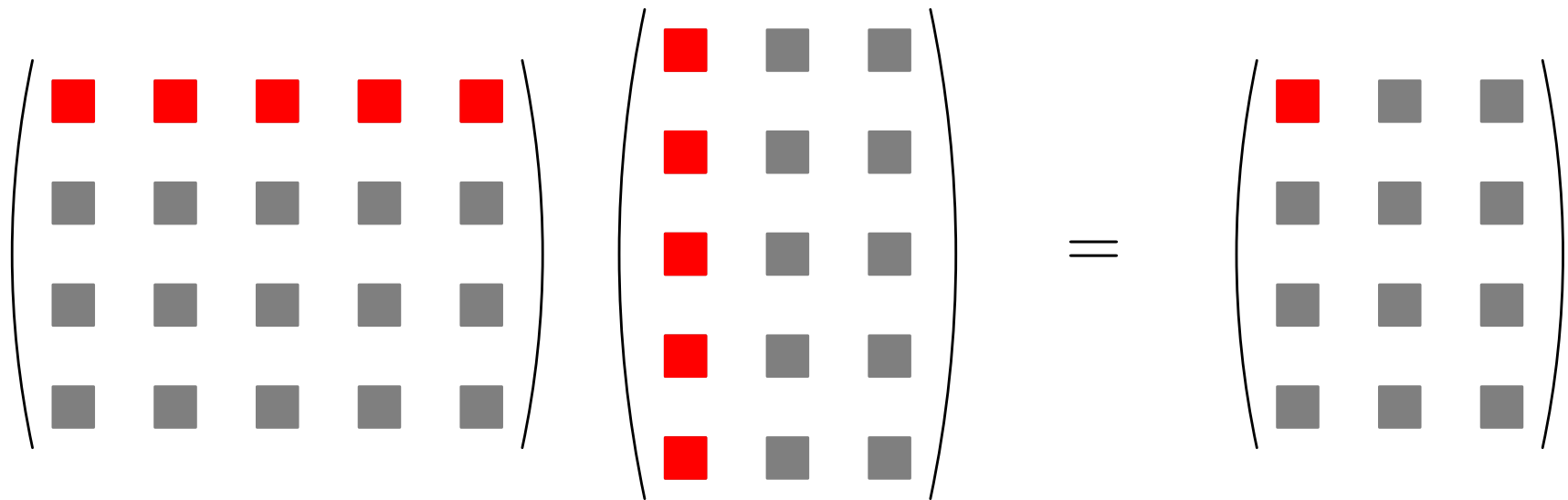
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The diagram illustrates matrix multiplication using a visual representation of matrices as grids of squares. The first matrix, representing $\mathbf{A}$, is a 5x5 grid with 5 rows and 5 columns. The second matrix, representing $\mathbf{B}$, is a 5x3 grid with 5 rows and 3 columns. An equals sign follows, leading to the resulting matrix $\mathbf{C}$, which is a 5x3 grid with 5 rows and 3 columns. This visualizes the operation $\mathbf{AB} = \mathbf{C}$.

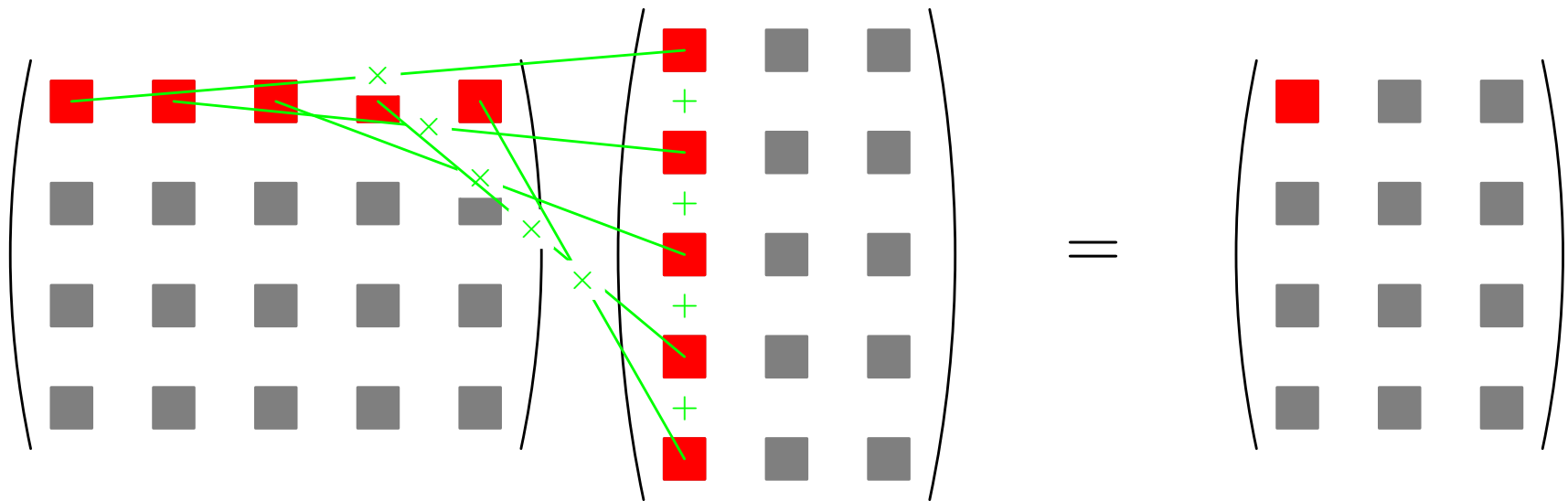
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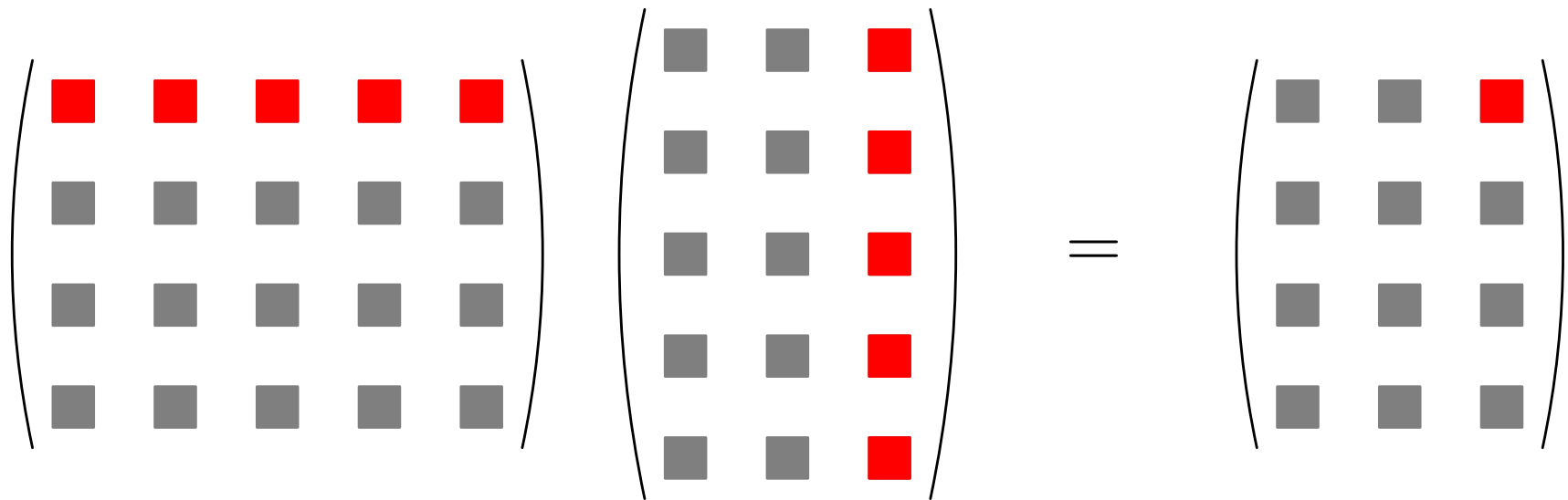
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The diagram illustrates the matrix multiplication $\mathbf{AB} = \mathbf{C}$ using a grid of squares. Matrix $\mathbf{A}$ is a 4×5 matrix with 5 red squares in the first row and 4 gray squares in each of the other three rows. Matrix $\mathbf{B}$ is a 5×3 matrix with 5 red squares in the second column and 4 gray squares in each of the other two columns. Matrix $\mathbf{C}$ is a 4×3 matrix with 1 red square in the second column and 4 gray squares in each of the other two columns. An equals sign is placed between matrix $\mathbf{B}$ and matrix $\mathbf{C}$.

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The diagram illustrates matrix multiplication using a grid of squares. The first matrix is 4 rows by 5 columns, with a red square in the second row, first column. The second matrix is 5 rows by 3 columns, with red squares in the first, second, third, fourth, and fifth rows, first column. The result is a 4 rows by 3 columns matrix, with a red square in the second row, first column.

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Gray	Gray	Gray	Gray	Gray	=	Gray	Gray	Gray
Red	Red	Red	Red	Red		Gray	Red	Gray
Gray	Gray	Gray	Gray	Gray		Gray	Gray	Gray
Gray	Gray	Gray	Gray	Gray		Gray	Gray	Gray

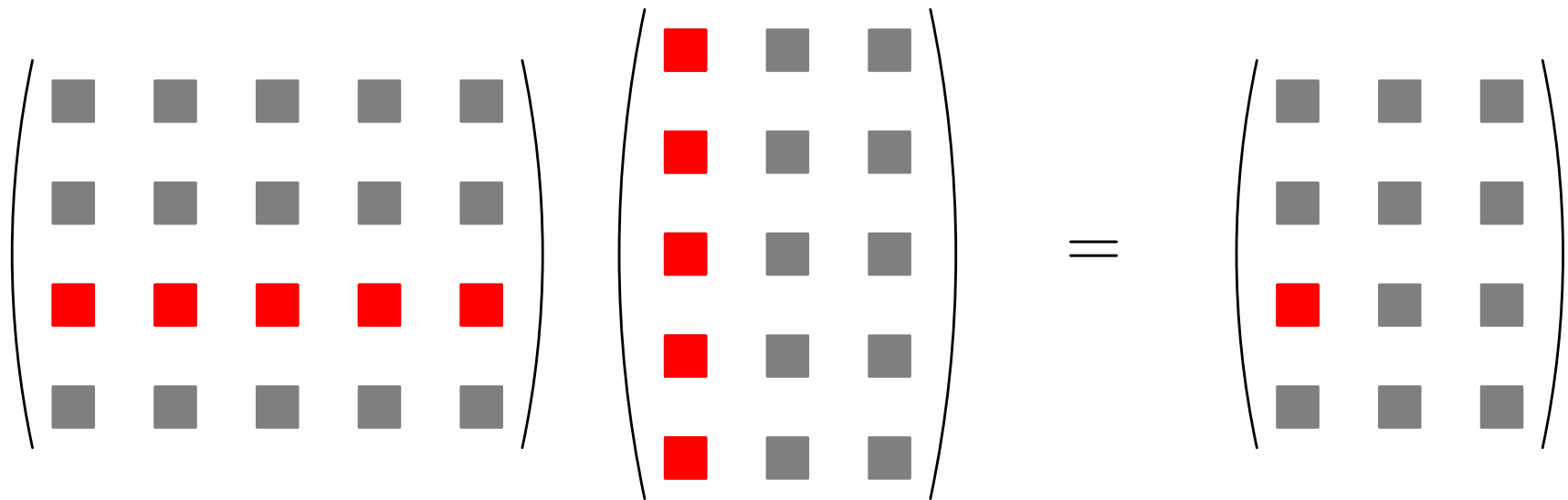
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The diagram illustrates the matrix multiplication $\mathbf{AB} = \mathbf{C}$. Matrix $\mathbf{A}$ is a 5x5 matrix with its third row highlighted in red. Matrix $\mathbf{B}$ is a 5x3 matrix with its second column highlighted in red. The result matrix $\mathbf{C}$ is a 5x3 matrix where only the third row is highlighted in red, representing the dot product of the third row of $\mathbf{A}$ and the columns of $\mathbf{B}$.

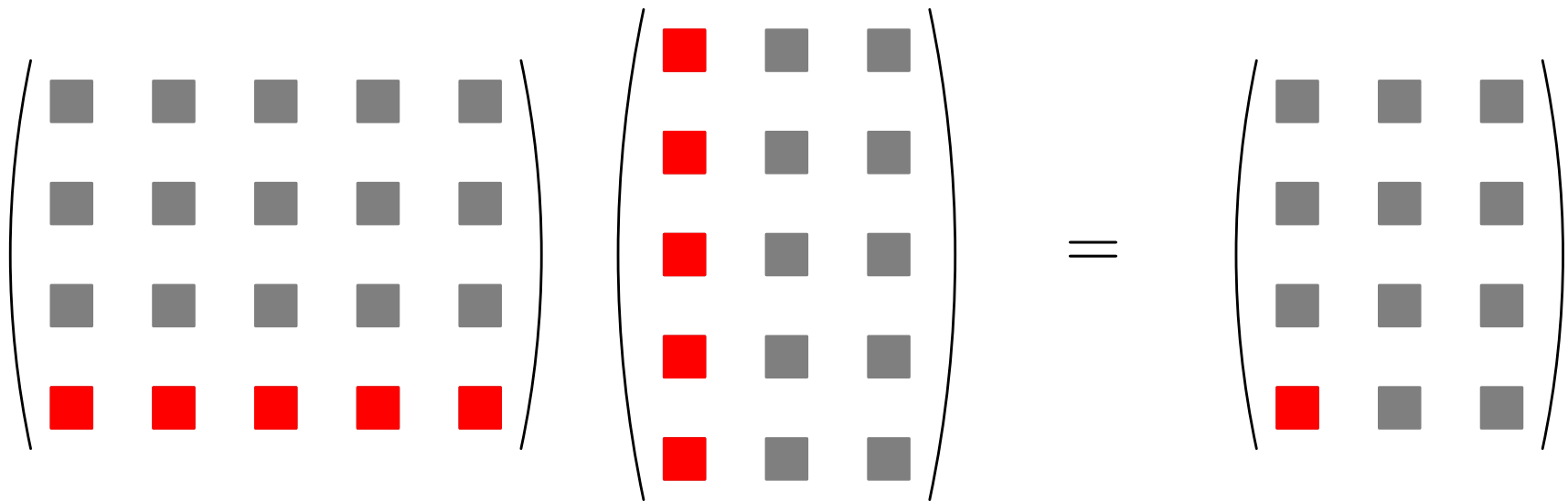
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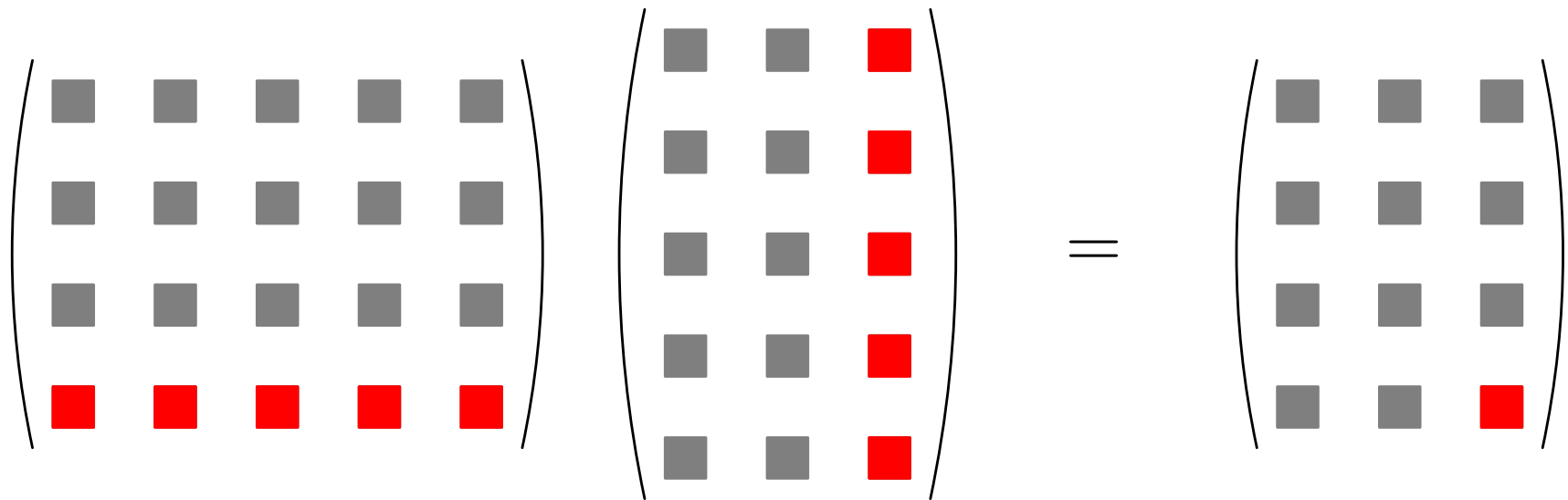
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The diagram illustrates matrix multiplication using a grid of squares. The first matrix is a 4x5 grid with 4 rows and 5 columns. The first three rows consist of 5 gray squares each. The fourth row consists of 5 red squares. The second matrix is a 5x3 grid with 5 rows and 3 columns. The first four rows consist of 2 gray squares followed by 1 red square. The fifth row consists of 2 gray squares followed by 1 red square. An equals sign follows, leading to the resulting 4x3 matrix. The first three rows consist of 3 gray squares each. The fourth row consists of 2 gray squares followed by 1 red square.

Matrix Multiplication

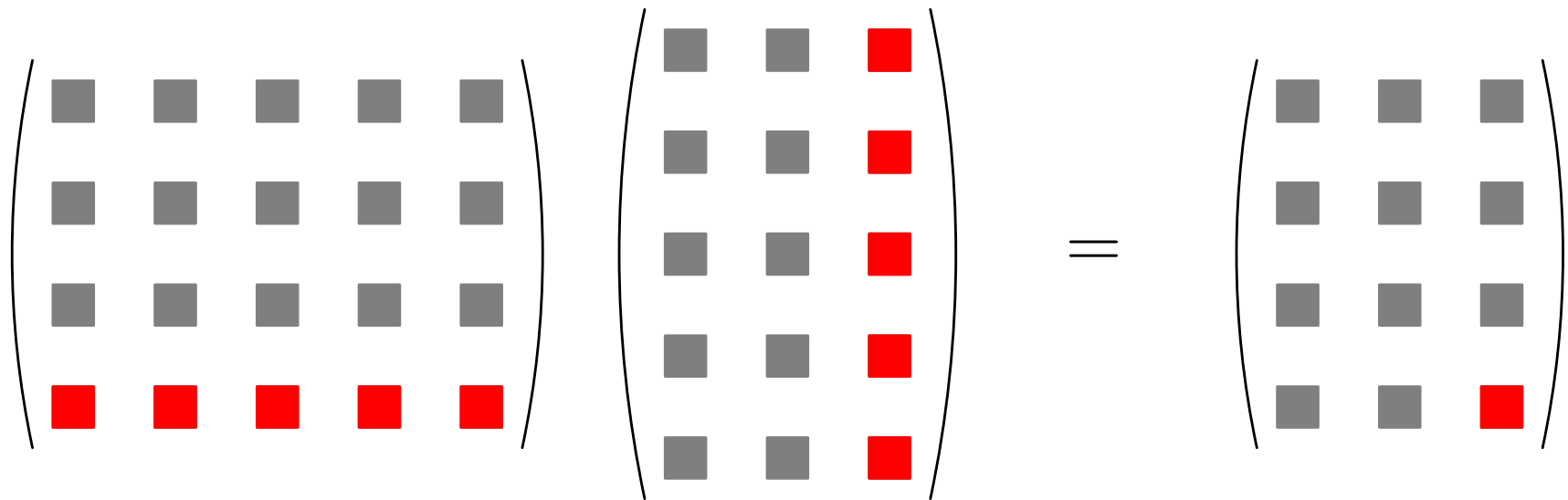
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- Matrix multiplication is associative: $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$

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- Matrix multiplication is associative: $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$
- But in general not commutative $\mathbf{AB} \neq \mathbf{BA}$

Transpose

- We often encounter transposes of a matrix $\mathbf{M}^T$ where we have exchanged the rows and columns: $[\mathbf{M}^T]_{ij} = [\mathbf{M}]_{ji}$
- Clear $(\mathbf{M}^T)^T = \mathbf{M}$
- It is simple question of book keeping to show that

$$(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$$

- For a symmetric matrix $\mathbf{M} = \mathbf{M}^T$
- Since 1×1 matrices are always symmetric

$$\mathbf{a}^T \mathbf{b} = (\mathbf{a}^T \mathbf{b})^T = \mathbf{b}^T \mathbf{a}$$

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$$\begin{aligned} \mathbf{a}^\top \mathbf{b} &= (\mathbf{a}^\top \mathbf{b})^\top = \mathbf{b}^\top \mathbf{a} \\ \mathbf{a}^\top \mathbf{M} \mathbf{b} &= \mathbf{b}^\top \mathbf{M}^\top \mathbf{a} \end{aligned}$$

Matrix-Vector Multiplication

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The diagram illustrates matrix-vector multiplication as matrix multiplication. On the left, a 4x5 matrix is represented by a large left parenthesis followed by a 4x5 grid of small gray squares, followed by a large right parenthesis. This is multiplied by a 5x1 column vector, represented by a large left parenthesis followed by a vertical column of five small gray squares, followed by a large right parenthesis. An equals sign follows, and then a 4x1 column vector, represented by a large left parenthesis followed by a vertical column of four small gray squares, followed by a large right parenthesis.

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The diagram illustrates matrix-vector multiplication as matrix multiplication. It shows a 4x5 matrix (with 5 red squares in the top row and 4 gray squares in each of the other three rows) multiplied by a 5x1 column vector (with 5 red squares). This is equated to a 4x1 column vector (with 1 red square in the top row and 3 gray squares in the other rows).

$$\begin{pmatrix} \text{red} & \text{red} & \text{red} & \text{red} & \text{red} \\ \text{gray} & \text{gray} & \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} & \text{gray} & \text{gray} \end{pmatrix} \begin{pmatrix} \text{red} \\ \text{red} \\ \text{red} \\ \text{red} \\ \text{red} \end{pmatrix} = \begin{pmatrix} \text{red} \\ \text{gray} \\ \text{gray} \\ \text{gray} \end{pmatrix}$$

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The diagram illustrates matrix-vector multiplication as matrix multiplication. It shows a 5x5 matrix (represented by a grid of squares, with the second row highlighted in red) multiplied by a 5x1 column vector (represented by a single column of squares, with all squares highlighted in red). The result is a 5x1 column vector (represented by a single column of squares, with the second square highlighted in red). The equation is represented as:

$$\begin{pmatrix} \text{gray} & \text{gray} & \text{gray} & \text{gray} & \text{gray} \\ \text{red} & \text{red} & \text{red} & \text{red} & \text{red} \\ \text{gray} & \text{gray} & \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} & \text{gray} & \text{gray} \end{pmatrix} \begin{pmatrix} \text{red} \\ \text{red} \\ \text{red} \\ \text{red} \\ \text{red} \end{pmatrix} = \begin{pmatrix} \text{gray} \\ \text{red} \\ \text{gray} \\ \text{gray} \\ \text{gray} \end{pmatrix}$$

Matrix-Vector Multiplication

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The diagram illustrates matrix-vector multiplication as matrix multiplication. On the left, a 5x5 matrix is shown with 4 gray squares and 1 red square in each row. This matrix is multiplied by a 5x1 column vector consisting of 5 red squares. The result is a 5x1 column vector consisting of 4 gray squares and 1 red square. The entire equation is enclosed in large parentheses.

$$\begin{pmatrix} \text{gray} & \text{gray} & \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} & \text{gray} & \text{gray} \\ \text{red} & \text{red} & \text{red} & \text{red} & \text{red} \\ \text{gray} & \text{gray} & \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} & \text{gray} & \text{gray} \end{pmatrix} \begin{pmatrix} \text{red} \\ \text{red} \\ \text{red} \\ \text{red} \\ \text{red} \end{pmatrix} = \begin{pmatrix} \text{gray} \\ \text{gray} \\ \text{red} \\ \text{gray} \\ \text{gray} \end{pmatrix}$$

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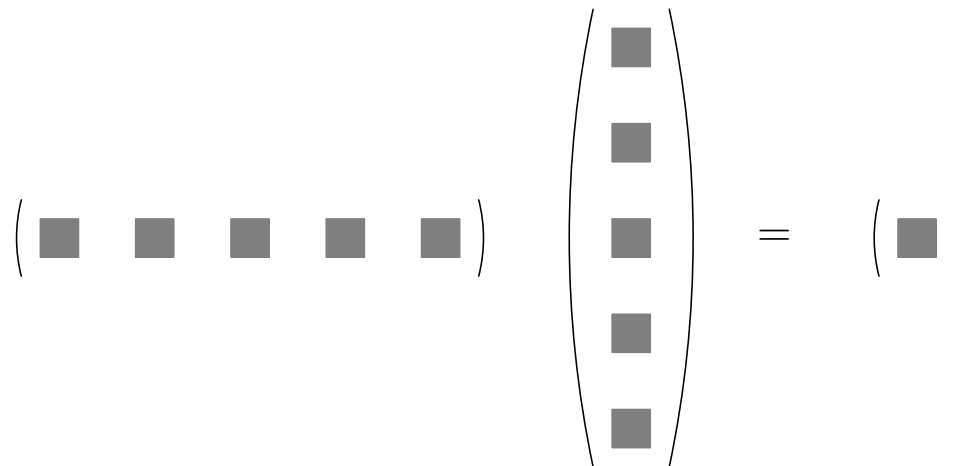
The diagram illustrates matrix-vector multiplication as matrix multiplication. It shows a 4x5 matrix (represented by a grid of 20 squares, with the bottom row highlighted in red) multiplied by a 5x1 column vector (represented by a single column of 5 red squares). The result is a 4x1 column vector (represented by a single column of 4 squares, with the bottom square highlighted in red). The entire operation is enclosed in large parentheses, with an equals sign indicating the result.

Inner Products

- The dot or inner-product between two vectors with the same number of elements, can be written $a^T b$

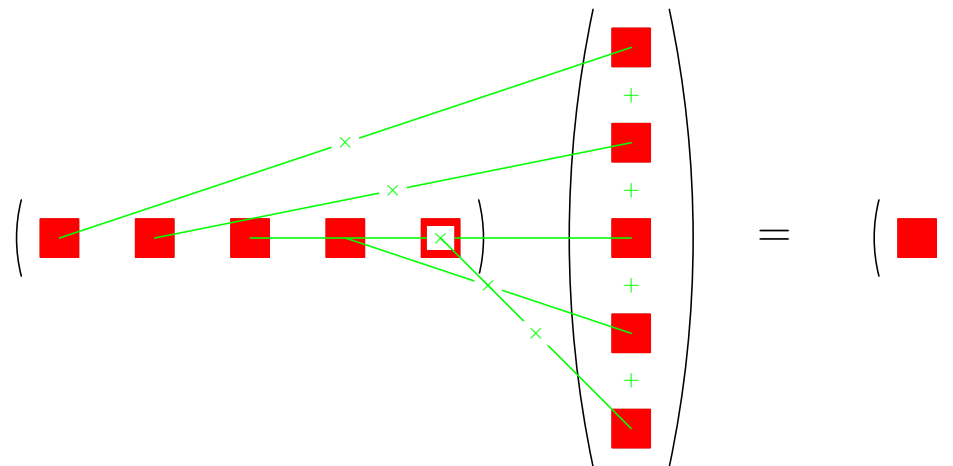
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$$\left(\begin{array}{ccccc} \blacksquare & \blacksquare & \blacksquare & \blacksquare & \blacksquare \end{array} \right) = \left(\begin{array}{c} \blacksquare \\ \blacksquare \\ \blacksquare \\ \blacksquare \\ \blacksquare \end{array} \right) = \left(\blacksquare \right)$$

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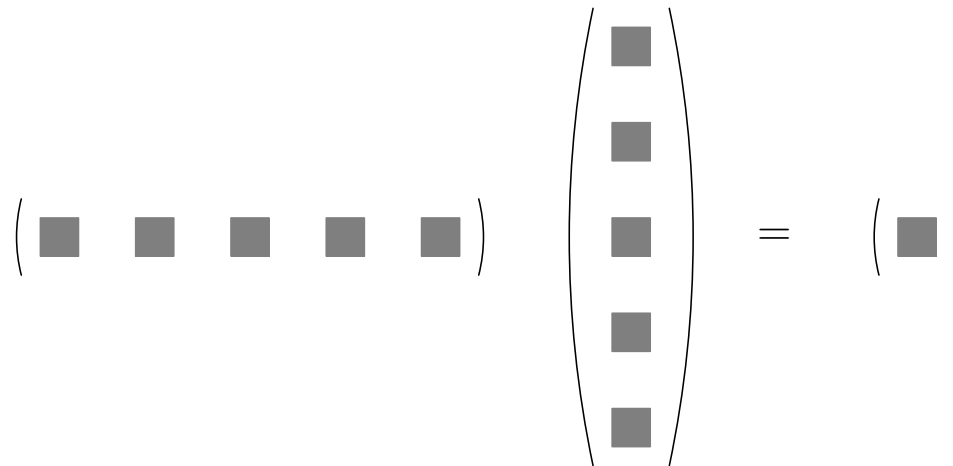
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- Inner products encode angles between vectors

$$\mathbf{a}^T \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos(\theta)$$

Outer Products

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The diagram illustrates the outer product operation ab^T . On the left, a column vector a is represented by a large left parenthesis followed by four gray squares stacked vertically, and a large right parenthesis. This is multiplied by a row vector b^T , represented by a large left parenthesis followed by three gray squares, and a large right parenthesis. An equals sign follows, leading to the resulting 4x3 matrix. This matrix is shown as a large left parenthesis followed by four rows, each containing three gray squares, and a large right parenthesis.

Outer Products

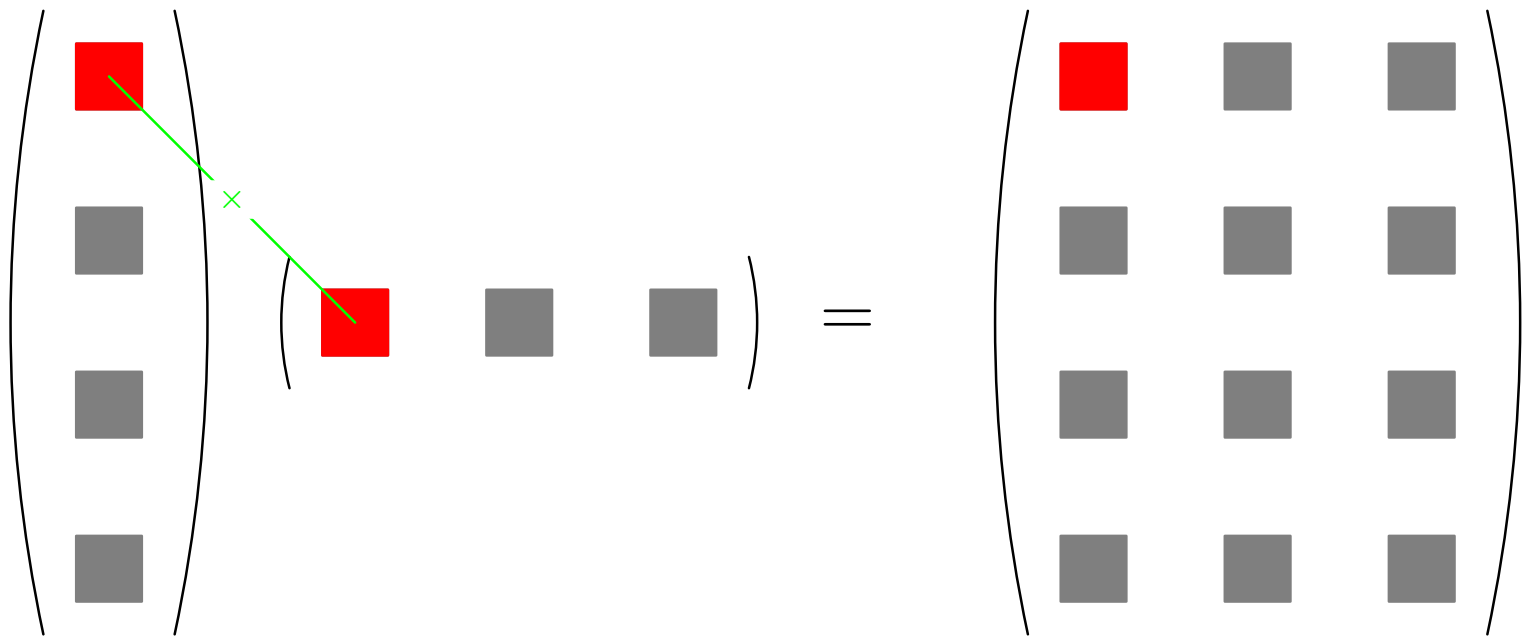
- We can also define the outer-product between any size vectors, ab^T

The diagram illustrates the outer product of two vectors. On the left, a column vector of size 4x1 is shown, with its top element highlighted in red and the others in gray. This is multiplied by a row vector of size 1x3, also with its first element highlighted in red and the others in gray. The result is a 4x3 matrix where each row is a scalar multiple of the original row vector, with the first row being the original row vector and the others being grayed-out copies.

$$\begin{pmatrix} \text{red} \\ \text{gray} \\ \text{gray} \\ \text{gray} \end{pmatrix} \begin{pmatrix} \text{red} & \text{gray} & \text{gray} \end{pmatrix} = \begin{pmatrix} \text{red} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} \end{pmatrix}$$

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Outer Products

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The diagram illustrates the outer product of two vectors. On the left, a column vector is shown with four elements: a red square at the top, followed by three gray squares. This is multiplied by a row vector with three elements: two gray squares followed by a red square. The result is a 4x3 matrix on the right, where each row is a scaled version of the row vector. The first row has a red square in the third column, while the other three rows have gray squares in all three columns.

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The diagram illustrates the outer product operation ab^T using square blocks to represent elements of matrices and vectors. On the left, a column vector a of size 4x1 is shown, with its second element highlighted in red. This is multiplied by a row vector b^T of size 1x3, with its first element highlighted in red. The result is a 4x3 matrix where each row is a scaled version of the vector b^T . The first row of the resulting matrix has its first element highlighted in red, corresponding to the red elements in the input vectors.

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The diagram illustrates the outer product operation ab^T using square blocks to represent elements of vectors and matrices. On the left, a column vector a is shown as a vertical stack of four squares, with the second square from the top highlighted in red. To its right is a row vector b^T shown as a horizontal row of three squares, with the middle square highlighted in red. An equals sign follows, leading to the resulting 4x3 matrix. This matrix is represented by a large set of parentheses containing a 4x3 grid of squares. The second row and third column of this grid are highlighted in red, showing that the red element from the column vector is multiplied by the red element from the row vector to produce the red element in the resulting matrix.

Outer Products

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The diagram illustrates the outer product operation ab^T . On the left, a column vector a is shown with four elements: three gray squares and one red square in the second position. This is multiplied by a row vector b^T with three elements: two gray squares and one red square in the third position. The result is a 4x3 matrix where each element is the product of the corresponding elements from a and b^T . The resulting matrix has four rows and three columns. The second row and third column contain the red square, representing the product of the red elements from the original vectors. All other elements are gray squares.

Outer Products

- We can also define the outer-product between any size vectors, ab^T

The diagram illustrates the outer product operation ab^T . On the left, a column vector a of size 4x1 is shown, with its third element highlighted in red. This is multiplied by a row vector b^T of size 1x3, with its first element highlighted in red. The result is a 4x3 matrix on the right, where each row is a scalar multiple of the row vector b^T , and the first row is identical to b^T because the red element in a is 1 (implied). The matrix has 4 rows and 3 columns, with the first row containing the red element and two gray elements, and the subsequent three rows containing gray elements.

Outer Products

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The diagram illustrates the outer product of two 4x1 vectors. On the left, a column vector of size 4x1 is shown, containing four gray squares, with the third square from the top highlighted in red. This is followed by a 1x4 row vector, also containing four gray squares, with the second square from the left highlighted in red. An equals sign follows, leading to a 4x4 matrix on the right. This matrix is the result of the outer product, where each element is the product of the corresponding elements from the two input vectors. The entire 4x4 matrix is enclosed in large parentheses. The element in the third row and second column is highlighted in red, corresponding to the red squares in the input vectors.

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The diagram illustrates the outer product operation ab^T . On the left, a column vector a is represented by a large left parenthesis followed by four gray squares, with the third square highlighted in red, and a large right parenthesis. This is multiplied by a row vector b^T , represented by a large left parenthesis followed by three squares (two gray, one red), and a large right parenthesis. An equals sign follows. On the right, the resulting 4x3 matrix is shown within large parentheses, consisting of four rows and three columns of squares. The third row and third column are highlighted in red, representing the outer product of the red squares from the original vectors.

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The diagram illustrates the outer product operation ab^T using square blocks to represent matrix elements. On the left, a column vector a of size 4x1 is shown, with its bottom element highlighted in red. This is multiplied by a row vector b^T of size 1x3, with its first element highlighted in red. The result is a 4x3 matrix on the right, where each row is a scalar multiple of the vector b^T . The bottom row of the resulting matrix has its first element highlighted in red, corresponding to the product of the red elements in the input vectors.

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The diagram illustrates the outer product of two vectors. On the left, a column vector a of size 4x1 is shown, with its bottom element highlighted in red. This is multiplied by a row vector b^T of size 1x3, with its middle element highlighted in red. The result is a 4x3 matrix on the right, where the bottom row is highlighted in red, representing the outer product ab^T .

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The diagram illustrates the outer product of two vectors. On the left, a column vector of size 4 is shown, consisting of three gray squares and one red square at the bottom. This is multiplied by a row vector of size 3, consisting of two gray squares and one red square on the right. The result is a 4x3 matrix on the right, where each element is the product of the corresponding elements from the two input vectors. The bottom-right element is red, while all other elements are gray.

$$\begin{pmatrix} \square \\ \square \\ \square \\ \color{red}\square \end{pmatrix} \begin{pmatrix} \square & \square & \color{red}\square \end{pmatrix} = \begin{pmatrix} \square & \square & \square \\ \square & \square & \square \\ \square & \square & \square \\ \square & \square & \color{red}\square \end{pmatrix}$$

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The diagram illustrates the outer product of a column vector and a row vector. On the left, a column vector is represented by a large left parenthesis followed by four squares stacked vertically, and a large right parenthesis. The bottom square is red, while the others are gray. This is followed by a row vector, represented by a large left parenthesis followed by three squares, and a large right parenthesis. The rightmost square is red, while the others are gray. An equals sign follows. To the right of the equals sign is a 4x3 matrix, represented by a large left parenthesis followed by four rows of three squares each, and a large right parenthesis. The bottom-right square (row 4, column 3) is red, while all other squares are gray.

- Notice that we need to know whether we are dealing with row or column vectors

Norms

- Norms are non-negative numbers associated with the “size” of a vector
- The norm you should be familiar with is the **Euclidean Norm** or 2-norm

$$\|\mathbf{x}\|_2 = \sqrt{\sum_{i=1}^n x_i^2}$$

- We will see there exist other norms (using different ways of measuring size)
- The Euclidean norm is intimately related to the inner-product, we will see other examples of this for different vector spaces

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Matrices as Linear Mappings

- Matrices represent mappings between vectors

$$\mathbf{b} = \mathbf{M}\mathbf{a}$$

- The elements of $\mathbf{b}$ are **linear sums** of the elements of $\mathbf{a}$

$$b_i = \sum_j M_{ij} a_j$$

- Two important consequences of this linearity are for any real c

$$\mathbf{M}(c\mathbf{a}) = c(\mathbf{M}\mathbf{a})$$

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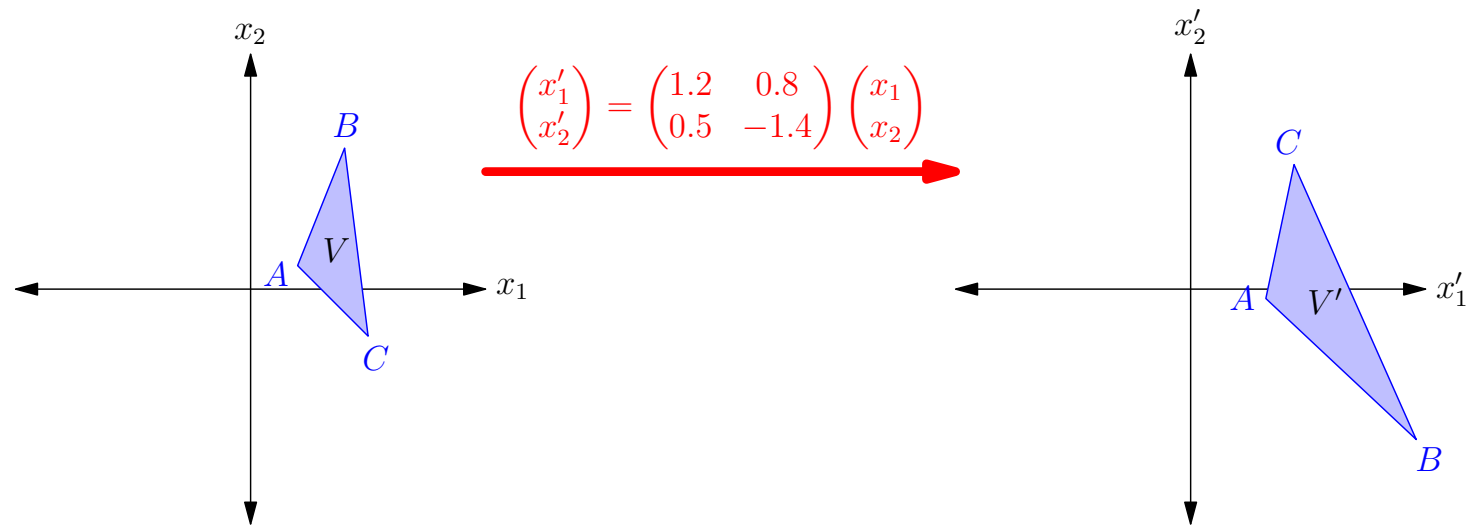
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$$\mathbf{M}(c\mathbf{a}) = c(\mathbf{M}\mathbf{a})$$

$$\mathbf{M}(\mathbf{a} + \mathbf{b}) = (\mathbf{M}\mathbf{a}) + (\mathbf{M}\mathbf{b})$$

Action Of Matrices on Volumes

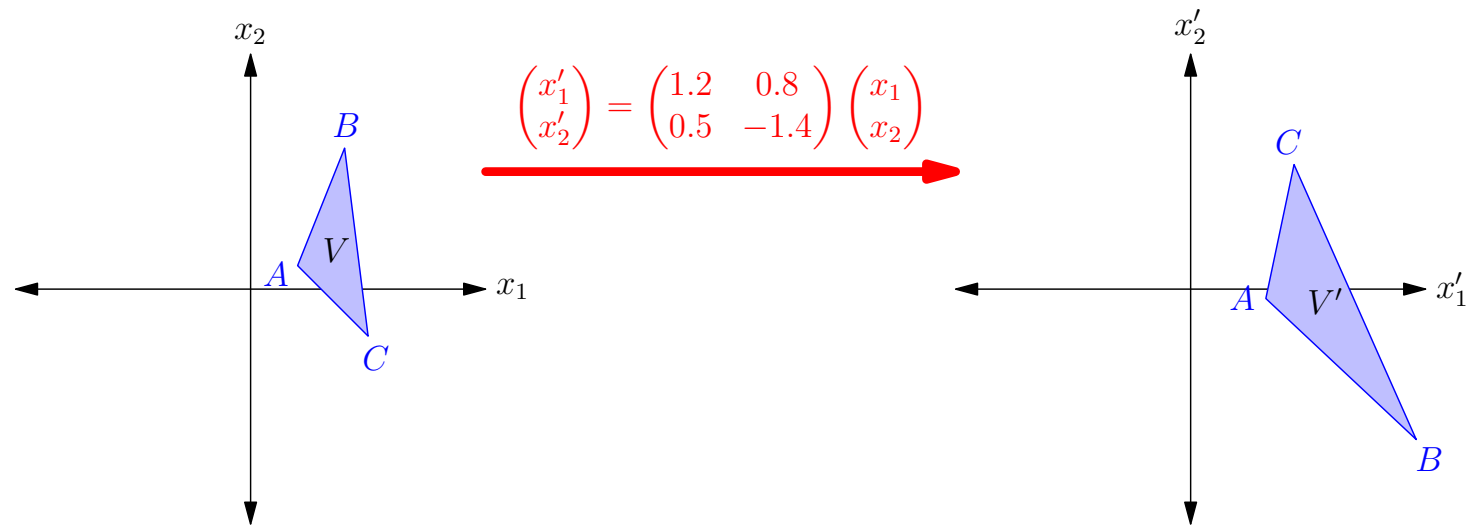
- For a closed set of points corresponding to a volume, V in $\mathbb{R}^n$, treating each point as a vector, we can ask what happens to the volume after mapping the vector by a matrix



- For square matrices, $\mathbf{M}$, the determinant is equal to $\det(\mathbf{M}) = \pm V'/V$
- The sign tells us if there has been a reflection

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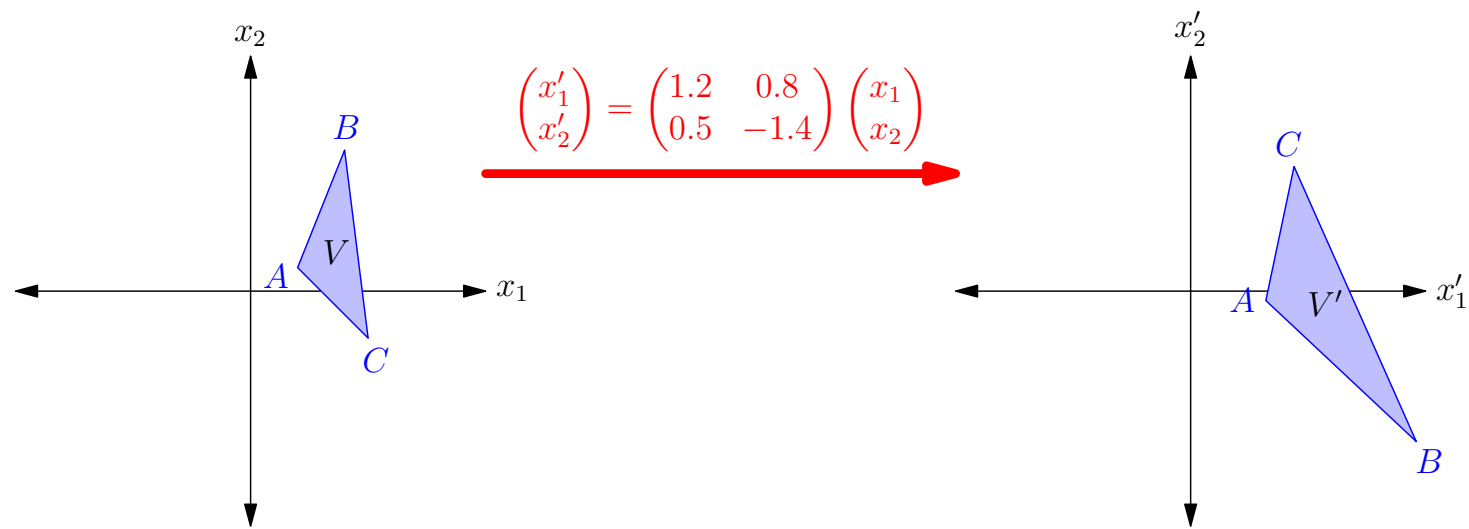
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Action Of Matrices on Volumes

- For a closed set of points corresponding to a volume, V in $\mathbb{R}^n$, treating each point as a vector, we can ask what happens to the volume after mapping the vector by a matrix



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Determinants

- A consequence of the fact that $\mathbf{AB}$ denotes the composition, apply mapping $\mathbf{B}$ then apply mapping $\mathbf{A}$

$$\det(\mathbf{AB}) = \det(\mathbf{A}) \times \det(\mathbf{B})$$

- Determinants are only defined for square matrices, as non-square matrices project into different size spaces, where the volumes cannot be compared
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Multivariate Functions

- In machine learning we heavily use multivariate functions

$$f : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

- Often our functions will be scalar value $f : \mathbb{R}^n \rightarrow \mathbb{R}$
- E.g. a learning machine for regression might be $\hat{f}(\mathbf{x}, \mathbf{w})$
- We say a function f is linear if

$$f(c\mathbf{x}) = cf(\mathbf{x}) \qquad f(\mathbf{x} + \mathbf{y}) = f(\mathbf{x}) + f(\mathbf{y})$$

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Gradients

- The gradient of a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ at a point $\mathbf{x}^*$ is defined as the column vector

$$\nabla f(\mathbf{x}) = \begin{pmatrix} \frac{\partial f(\mathbf{x}^*)}{\partial x_1} \\ \frac{\partial f(\mathbf{x}^*)}{\partial x_2} \\ \vdots \\ \frac{\partial f(\mathbf{x})^*}{\partial x_n} \end{pmatrix}$$

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- To reason about events with uncertain outcomes we introduce **random variables**
- Random variables corresponds to objects that we assign numbers to depending on the outcome of a random event
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- A continuous random variable might be the temperature at 12 noon tomorrow in Hyde park

Probability Distributions

- For discrete events, we can define a **probability mass function** $\mathbb{P}(X = x)$ that the random variable takes the value x
- This is a number between 0 and 1 with $\sum_{x \in \mathcal{X}} \mathbb{P}(X = x) = 1$
- For continuous events the probability of X taking a particular value x is zero
- Instead we consider the probability density $f_X(x)$ which we can define as

$$f_X(x) = \lim_{\delta x \rightarrow 0} \frac{\mathbb{P}(x < X \leq x + \delta x)}{\delta x}$$

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Expectations

- One of the most useful quantities in dealing with random numbers is to look at the average value, aka as the **expectation**, for some function $g(X)$ of the random variable

$$\mathbb{E}[g(X)] = \begin{cases} \sum_{x \in \mathcal{X}} \mathbb{P}(X = x) g(x) \\ \int f_X(x) g(x) dx \end{cases}$$

- Expectations are linear operators

$$\mathbb{E}[cg(X)] = c\mathbb{E}[g(x)]$$

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- We will revisit most of these topics to gain a better understanding, but you will be lost in the other machine learning modules if you are not familiar with everything I've covered
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